IsoLine and Gaussian Emitters

Sean Wang | May 2026

Directional variation outperforms random mutation on a relatively simple quality-diversity search.

1 Motivation

Emitters are search operators in Multi-dimensional Archive of Phenotypic Elites (MAP-Elites) [1,2] — an essential algorithm for quality-diversity (QD) optimization [3]. While augmented emitters are known to improve searches on complex fitness landscapes [4–6], we depart from this consensus to instead explore the relevance of emitter augmentation on simpler terrains. The directional variation (IsoLine) emitter is our sample subject [6], the random mutation (Gaussian) emitter is our baseline control [2],[1] and a deeply constrained stochastic Lindenmayer system (SL-System) provides our fitness landscape [7]. More specifically, we test if the IsoLine emitter produces a higher quality-diversity score (QD-score) after the same total searches [3].

In this paper, we begin by conceptualizing our emitters and SL-system along with its measures. We then execute our searches to compare emitter performances. Finally, we revisit the notion of complexity by extending our insights towards open-ended evolution [8].

2 Concepts

Our exposition here is brief. Readers are directed to the Resources section for additional context.

2.1 Emitters

Gaussian emitter $\mathbf{E} : \mathcal{A} \mapsto \dot{\mathbf{x}}_\mathrm{i}$ samples $\mathbf{x}_\mathrm{i}$ from elites $\mathcal{A} \subset \mathbb{X}$ such that $\mathbb{X}$ is our search space [2]. The emitter then transforms $\mathbf{x}_\mathrm{i}$ to $\dot{\mathbf{x}}_\mathrm{i}$ by an isotropic Gaussian noise $\sigma_\mathrm{iso}\mathcal{N}(\mathbf{0}, \mathbf{I})$ such that σ_iso is a constant scalar, and $\mathbf{I} \in \mathbb{X}$ is the identity covariance matrix [6]. Furthermore, IsoLine emitter $\hat{\mathbf{E}} : \mathcal{A} \mapsto \ddot{\mathbf{x}}_\mathrm{i}$ further transforms $\dot{\mathbf{x}}_\mathrm{i}$ to $\ddot{\mathbf{x}}_\mathrm{i}$ via directional noise $\sigma_\mathrm{line}\mathcal{N}(0, 1)(\mathbf{x}_\mathrm{j} - \mathbf{x}_\mathrm{i})$ such that σ_line is another constant scalar, and $\mathbf{x}_\mathrm{j} \in \mathcal{A}$ is another sample elite [6]. We denote our emitters as

$$\hat{\mathbf{E}}(\mathcal{A}) := \dot{\mathbf{x}}_\mathrm{i} + \sigma_\mathrm{line}\mathcal{N}(0, 1)(\mathbf{x}_\mathrm{j} - \mathbf{x}_\mathrm{i}) = \ddot{\mathbf{x}}_\mathrm{i} \quad : \quad \mathbf{E}(\mathcal{A}) := \mathbf{x}_\mathrm{i} + \sigma_\mathrm{iso}\mathcal{N}(\mathbf{0}, \mathbf{I}) = \dot{\mathbf{x}}_\mathrm{i} \tag{1}$$

to establish IsoLine as an augmentation upon Gaussian [6]. The former can outsearch the latter by exploiting correlations between elites, which almost always cluster in the search space [6].

2.2 SL-System

SL-System $\mathcal{G} : \mathbf{x} \mapsto \mathcal{L}$ composes branching geometry $\mathcal{L}$ from a fixed set of botanic operations [7]. The system initiates from a fixed origin and develops under a fixed set of 5 rules with varying weights $\mathbf{x} \in \mathbb{X}$ such that $\mathbb{X} \subseteq \mathbb{R}^5$ [7]. Moreover, fitness function $F : \mathcal{L} \mapsto \mathbb{R}$ measures structural balance [2,3], while feature function $\mathbf{B} : \mathcal{L} \mapsto \mathbb{R}^2$ measures branch density and aspect stature [2,3]. We normalize all measures for each $\mathbf{x}$ across numerous compositions as $\bar{F}[\mathcal{G}(\mathbf{x})]$ and $\bar{\mathbf{B}}[\mathcal{G}(\mathbf{x})]$.

[1] We deliberately adopt the terms "subject" and "control" to illuminate the biological inspirations of QD optimization.

3 Execution

We conduct 50 comparison trials for a statistically significant sample size. During each trial, our emitters $\hat{\mathbf{E}}$ and $\mathbf{E}$ each conduct 500 searches with 50 new candidates per search [1], such that our 50×50 grid archive receives an ample supply of candidates [2]. The initial weights $\mathbf{x}_0 \in \mathbb{X}$ follow a constant uniform distribution [1,7], while $(\sigma_{\text{iso}}, \sigma_{\text{line}}) := (0.1, 0.2)$ to emphasize the directionality of $\hat{\mathbf{E}}$ [6]. In addition, the horizontal and vertical grid axes respectively represent branch density and aspect stature on intervals $[0.0, 1.5]$ and $[0.0, 8.0]$ [1,2]; we estimate these intervals as sufficiently large such that the archive can capture all $\mathbf{x} \in \mathbb{X}$ whose corresponding $\bar{\mathbf{B}}[\mathcal{G}(\mathbf{x})]$ is finite [1,2]. At the end of each 500 searches, we record the QD-score as $\sum_{\mathbf{x}\in\mathcal{A}} \bar{F}(\mathcal{G}(\mathbf{x}))$ such that $\bar{F}$ normalizes on interval $[0.0, 1.0]$ for all $\mathbf{x} \in \mathcal{A}$ [2,3] .

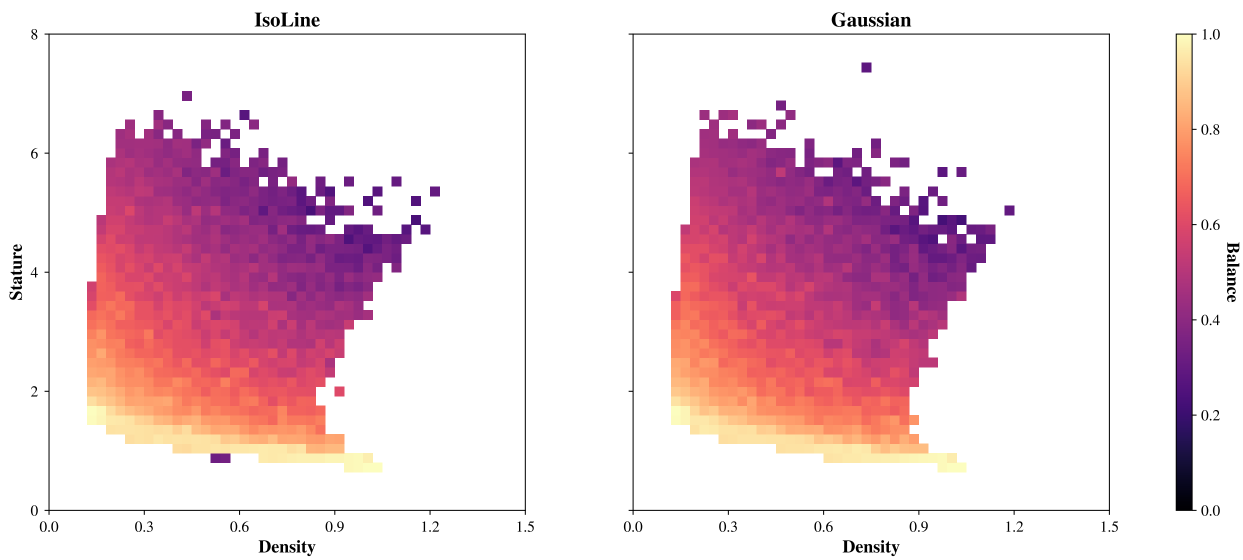

Figure 1: Archives from Trial 25 (subject QD-score = 493.7 | control QD-score = 492.5).

In the final analysis, the IsoLine emitter produces a higher QD-score after the same total searches. The subject QD-scores average at 493.96 ± 2.74, while those of the control do so at 492.11 ± 4.00. Moreover, the corresponding significance and effect size are respectively $(t(50), p) = (2.70, 0.01)$ and $g = 0.54$. We conjecture that our search space is sufficiently rich in clustering such that directional exploitation offers a clear advantage over canonical explorations [6, 7]. Thus, we conclude that emitter augmentation can improve searches even on simpler terrains. Finally, the search trajectories of our emitters remain an uncharted area worth exploring [4–6].

4 Extension

Like the majority of optimization heuristics [9], our emitters operate in a closed environment [8]. They rely on static measures under fixed descriptions [2,3], but many real-world systems do not conform to such rigidity [8]. Instead, they evolve novel ways to solve novel problems [8–10]. This open-endedness distinguishes complexity from mere complicatedness [8], yet our conceptions of the *complex* still revolve mostly around the latter [8]. Thus, our approach to MAP-Elites, QD, and optimization at large, can further improve through increasingly more dynamic representations of the fitness landscape [8,9] — we wish to develop beacons illuminating not merely interiors of the search space [2,3], but the very ontology of its adjacent possible [10].

5 Resources

The full implementation script is available at github.com/seanwang50/isoline-and-gaussian-emitters.

---

[1] Bryon Tjanaka, Matthew C. Fontaine, David H. Lee, Yulun Zhang, Nivedit Reddy Balam, Nathaniel Dennler, Sujay S. Garlanka, Nikitas Dimitri Klapsis, and Stefanos Nikolaidis. pyribs: A bare-bones python library for quality diversity optimization, 2023.

[2] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *CoRR*, abs/1504.04909, 2015.

[3] Justin K. Pugh, Lisa B. Soros, and Kenneth O. Stanley. Quality diversity: A new frontier for evolutionary computation. *Frontiers in Robotics and AI*, Volume 3 - 2016, 2016.

[4] Matthew C. Fontaine, Julian Togelius, Stefanos Nikolaidis, and Amy K. Hoover. Covariance matrix adaptation for the rapid illumination of behavior space. *CoRR*, abs/1912.02400, 2019.

[5] Matthew C. Fontaine and Stefanos Nikolaidis. Differentiable quality diversity. *CoRR*, abs/2106.03894, 2021.

[6] Vassilis Vassiliades and Jean-Baptiste Mouret. Discovering the elite hypervolume by leveraging interspecies correlation. *CoRR*, abs/1804.03906, 2018.

[7] Jason Bernard and Ian McQuillan. Stochastic l-system inference from multiple string sequence inputs, 2020.

[8] Wolfgang Banzhaf, Bert Baumgaertner, Guillaume Beslon, René Doursat, James A. Foster, Barry McMullin, Vinicius Veloso De Melo, Thomas Miconi, Lee Morrison, Michael Schanz, Moshe Sipper, Andrew Stout, and Elio Tuci. Defining and simulating open-ended novelty: requirements, guidelines, and challenges. *Theory in Biosciences*, 135(3):131–161, 2016.

[9] Rui Wang, Joel Lehman, Jeff Clune, and Kenneth O. Stanley. Paired open-ended trailblazer (POET): endlessly generating increasingly complex and diverse learning environments and their solutions. *CoRR*, abs/1901.01753, 2019.

[10] Marina Cortês, Stuart A. Kauffman, Andrew R. Liddle, and Lee Smolin. The tap equation: Evaluating combinatorial innovation. *European Economic Review*, 179:105144, October 2025.